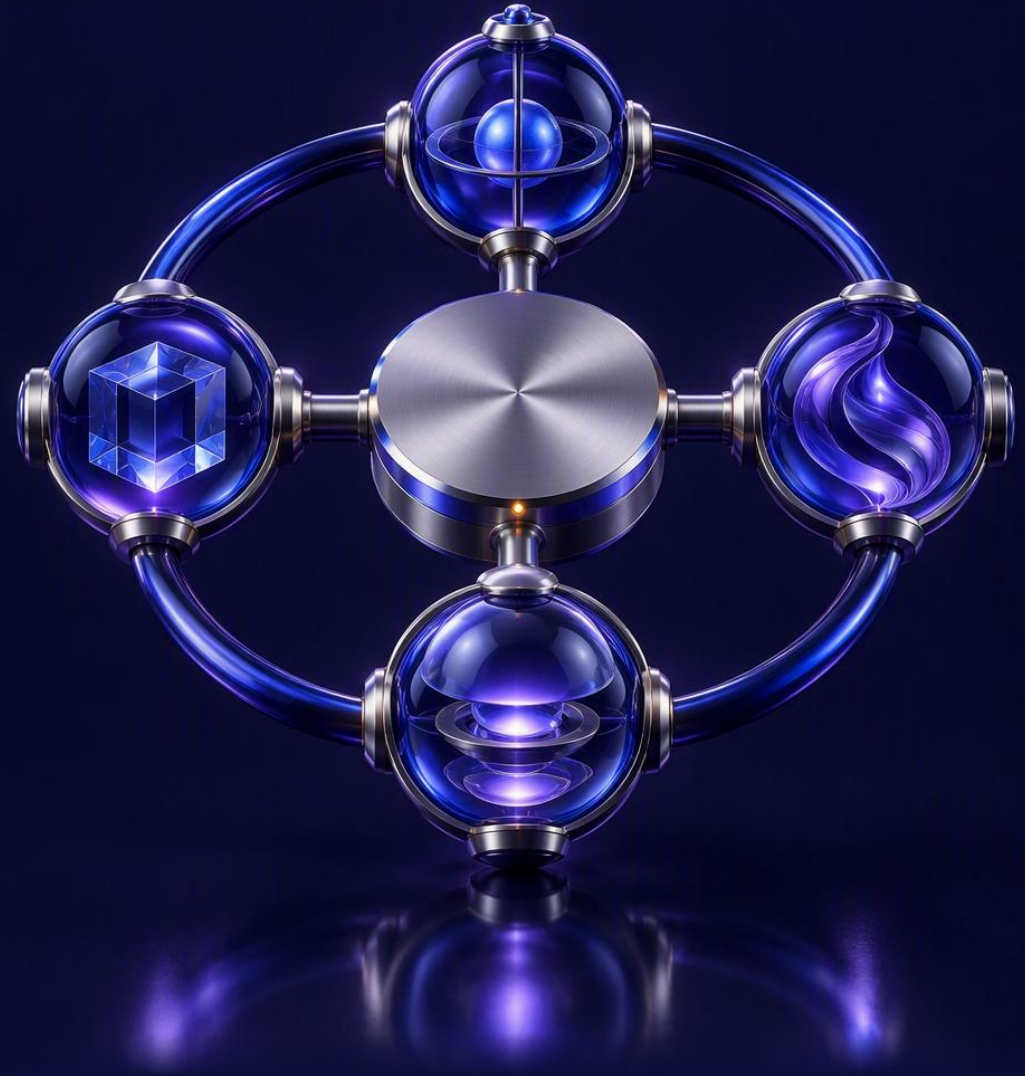


From tools to workflows

Skills, Shell, CodeAct
and Background Agents

Mock data. Real boundaries.

Bruno Capuano



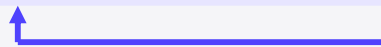
What is an agent?

An agent uses a model to decide the next step toward a goal.

Instructions	Purpose
Context	Information to work with
Tools	Actions, when needed



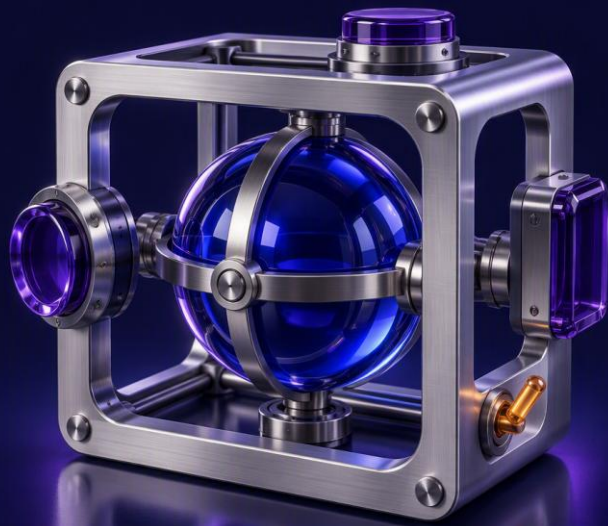
Decide → Act if needed → Observe → Answer



Continue when needed

Why do we need a harness?

Useful agents need more than a good prompt.



1

Run the loop

Model calls, tools and results

2

Maintain context

Conversation, plans and memory

3

Control execution

Configured approvals and policies

The model requests. The application enforces.

MAF supplies the harness.

We supply the application.

Microsoft Agent Framework

Model / tool loop
Session + context
Capability providers

Our application

Purpose + instructions
Tools + permitted data
Approvals + limits

`chatClient.AsHarnessAgent(...)`

Less plumbing. Explicit responsibility.



What have we built so far?

Same finance assistant.
More capability at each step.



PREVIOUSLY

SESSION 1

Agent to harness
Tools + web search
Planning + todos



PREVIOUSLY

SESSION 2

Scoped file access
Approval-gated actions
Durable memory



TODAY

SESSION 3

Discoverable skills
Approved shell actions
CodeAct + background
agents

More capability. Same explicit boundaries.

Start with one useful answer.

10 / PLAIN C#

Open the skill



SKILL.md

mock-prices.csv

25 shares at

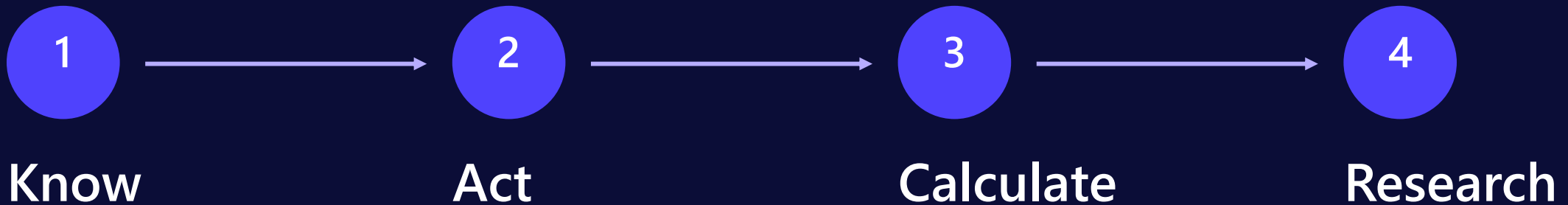
\$425

=

\$10,625

An inspectable package. A checkable result.

A morning brief is a workflow.



One goal. Four bounded capabilities.

Skills package what the agent needs.

Discover	Names and descriptions
Load	Selected instructions
Use	Bundled reference data

Loading a skill does not grant execution permission.



Same package. Less host plumbing.

10 / PLAIN C#

**The host
selects a package.**



11 / MAF BRIDGE

**The provider loads
on demand.**

AgentSkillsProviderBuilder

UseFileSkills(...)

AIContextProviders

Host-owned folder. Scripts denied.



11 / LIVE FOUNDRY

Let the agent
find the skill.

Value 25 shares of MSFT
using the mock data.

Expected fixture result

\$10,625

Observe the load. Then inspect the answer.

A request is not permission.

Request	Inspect the proposed action
Approve	A visible human decision
Execute	Only within host policy

LocalShellExecutor + approval-aware Harness





Tidy files. Keep the decision.

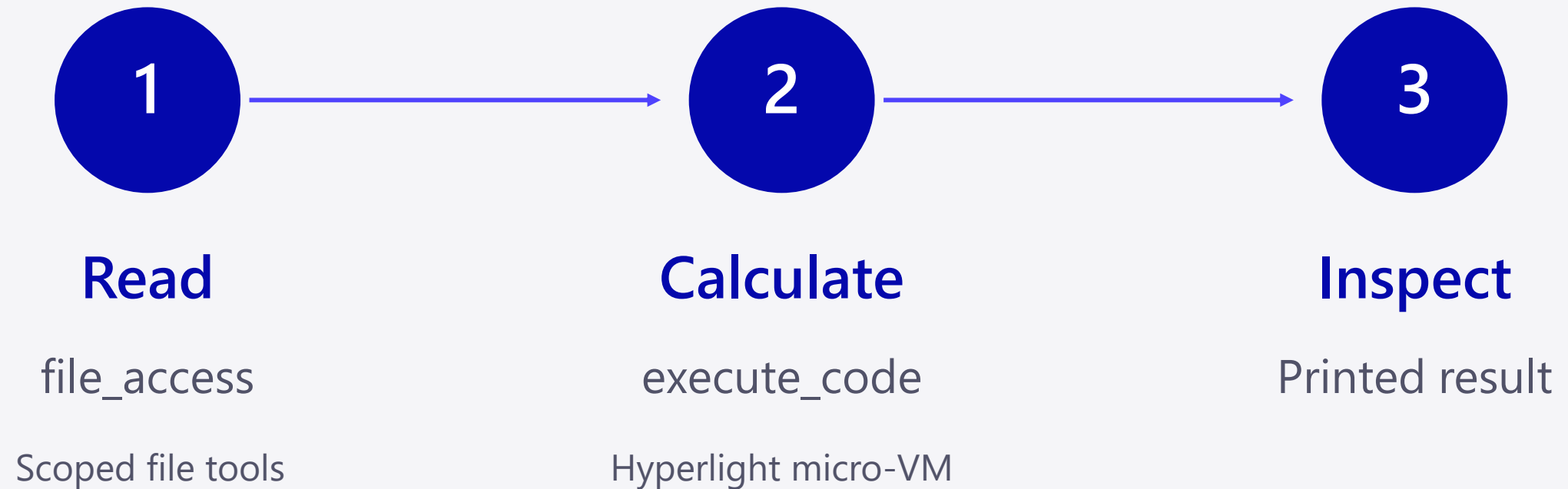
Tidy up my
trade confirmations.

`working\confirmations`

15-second timeout

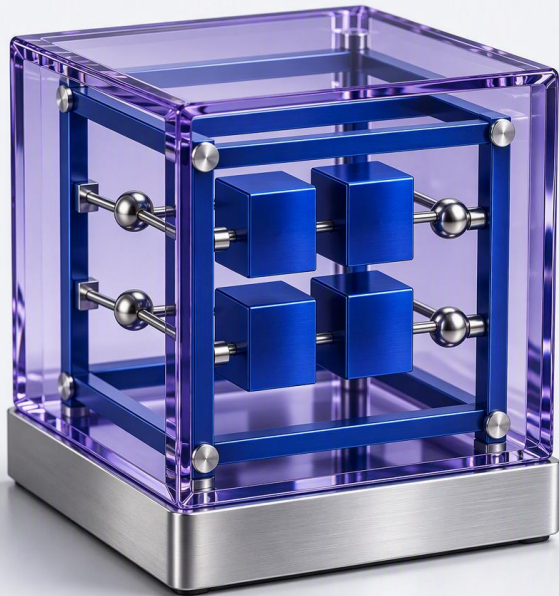
Denied means not executed.

Read. Calculate. Inspect.



Approval before generated code runs.

Compute the answer. Show the working.



31 / LIVE FOUNDRY

\$27,124.95

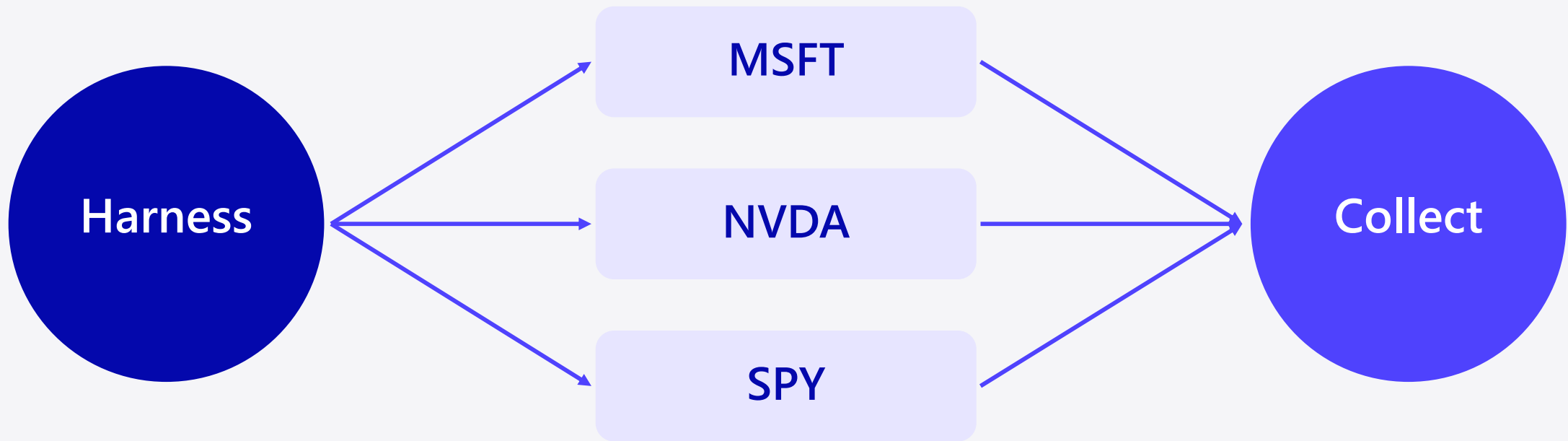
Portfolio total

66.01%

Technology

Expected fixture result. Four rows. Synthetic prices.

Delegate a question, not your authority.



One focused research agent

HostedWebSearchTool only

A dispatched task is not a completed finding.

41 / LIVE FOUNDRY

Three tickers. Three research lanes.

MSFT / NVDA / SPY

Collect cited findings.

Live public news. Variable results.



Compose what the host can prove.

OFFLINE ADVISOR / ONE RUN

1

Inspect

Mock portfolio

2

Check

dotnet --version

3

Queue

Follow-up ticket

Host-driven composition, not all four live bridges combined.

Keep status honest.

Completed

`shell.exitCode: 0`

An observed process result

Queued

`backgroundTask.status: queued`

A ticket, not completed research

No durable completion guarantee.

Four controls stay with the host.



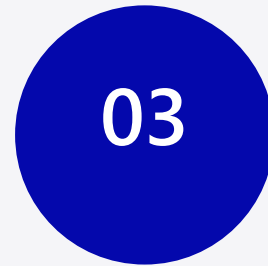
Scope

Packages and tools



Approval

Side effects and code



Isolation

Execution boundaries

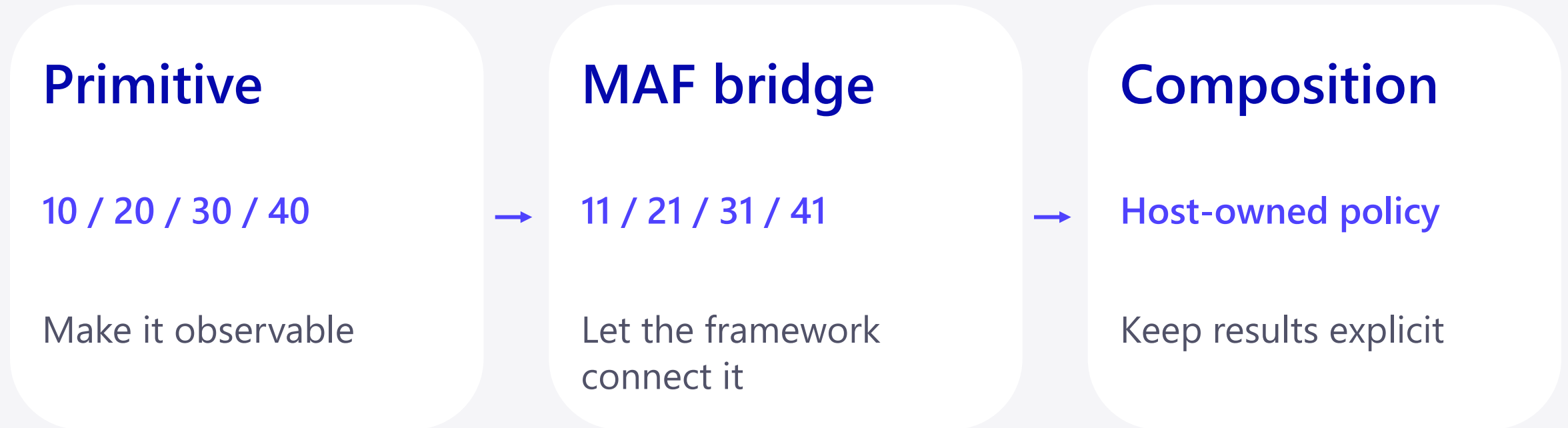


Lifecycle

Status and ownership

The model may request. The application decides.

Build one bounded workflow.



Capability grows. Responsibility stays explicit.

When live is unavailable, keep the evidence.

11 to 10

Local skill package

21 to 20

Fixed-command boundary, not cleanup

31 to 30

Two-row arithmetic, not the micro-VM

41 to 40

Ticket only, not completed research

Offline proof. Never simulated success.

More capability. Not more permission.

What will your next
bounded workflow do?

aka.ms/mafclaw/repo

Bruno Capuano

Questions

